BMS COLLEGE OF ENGINEERING, BANGALORE-19

(Autonomous Institute, Affiliated to VTU)

Department of Electronics and Communication

INTERNAL-I

Course Code : 23EC3ESHDL	Course Title : HDL Programming		
Semester : III	Maximum Marks: 40	Date: 04-01-2024	Time: 12:00 – 1:15pm
Faculty Handling the Course: Dr. RJG, Dr. SP, RCR,JDR, PJ	Sections: A,B,C,D and E		

PART-I (REMEMBER/UNDERSTAND)

No.	Question	Marks	CO	PO
1.	Explain the VLSI design flow with the help of flow diagram.	10	-	-

PART-III (DESIGN/ANALYZE/ENGINEERING TOOL/ANY OTHER)

No.	Questions **Note: Write the required logic/block diagram and truth table (for each question)**	Marks	CO	PO
2	Design 4-bit Ripple Carry Adder(RCA) using Verilog top-down design methodology. Also write the test code to test the functioning of the designed RCA.	10	3	3
3	Design a 4x1 Multiplexer using 2x1 multiplexer using Verilog. Also implement the 2x1 multiplexer using Verilog gate-level descriptions.	10	3	3
4	Design the logic circuit for the expression F = wxy' + y'z + w'yz'+x'yz' and implement the design using Verilog gate level modelling by considering the delay for the gates as given, NOT gate = 2ns, AND gate = 4ns, OR gate = 3ns. Write test bench code for the input values of w=1, x= 0, y=1, z=1 and w=0, x= 1, y=0, z=1. Draw the simulated waveform as per the given input.	10	3	3

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